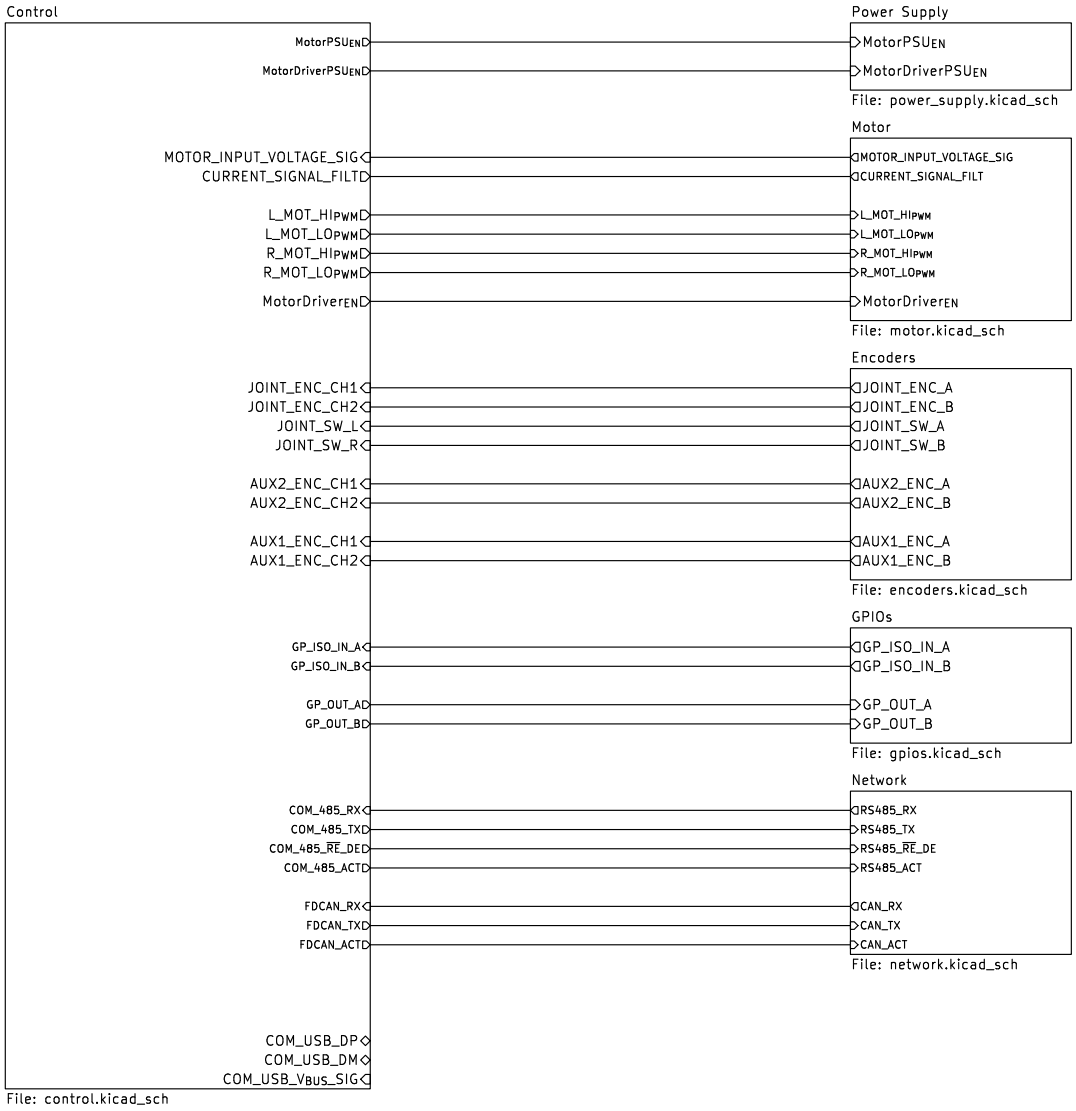
System overview



H1 H2
H3 H4

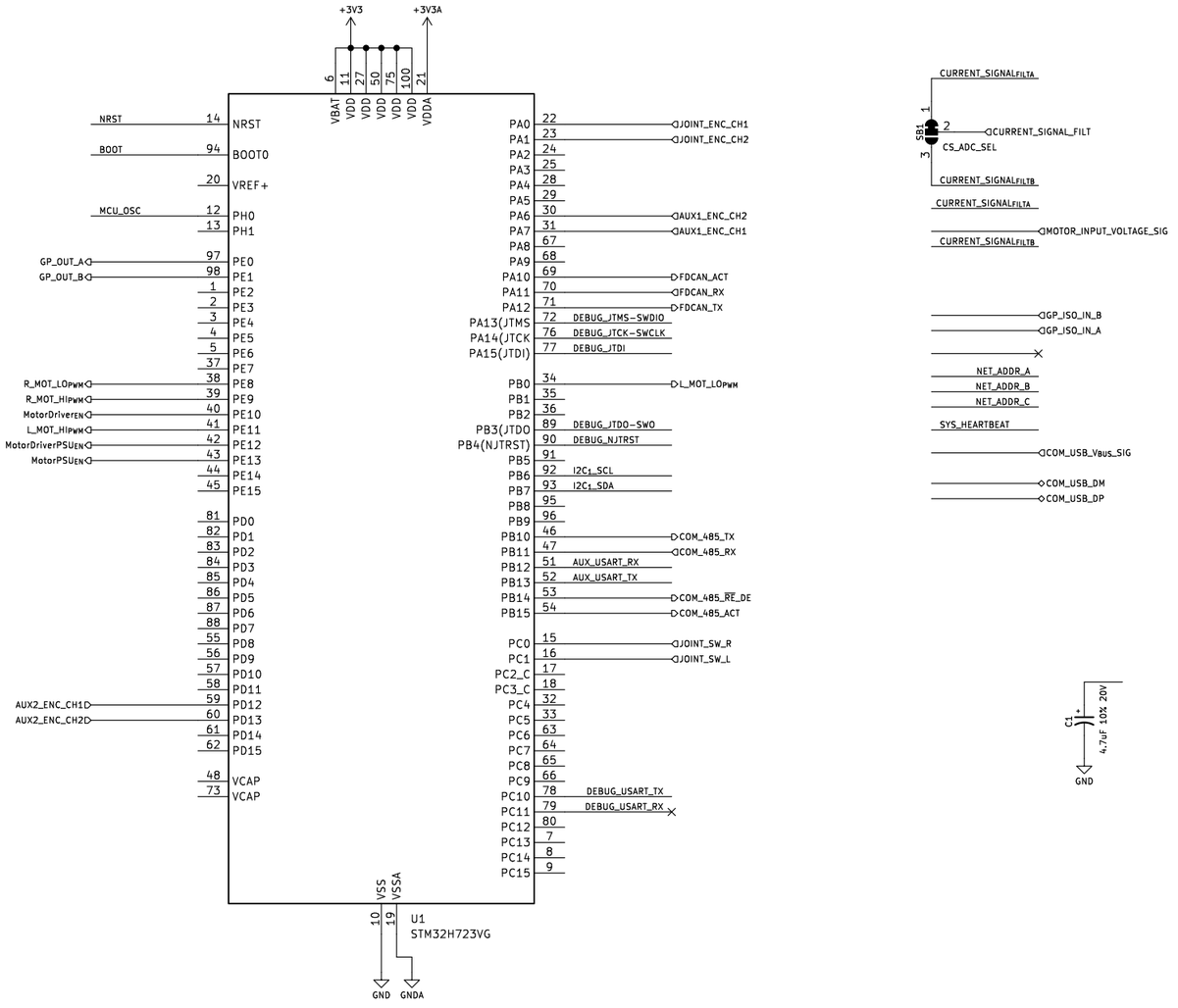
Single Joint Controller Board
Designed by Alves, T. G. (tgiacomellibr)

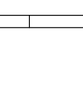
Sheet: /
File: hardware.kicad_sch

Title: System overview

Size: A4	Date: 2024-05-06	**Rev: 1.0**
KiCad E.D.A. 8.0.8		Id: 1/10

Microcontroller

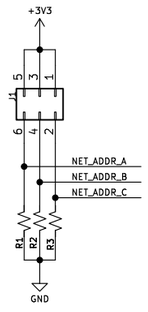




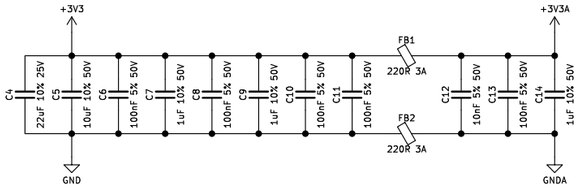
Network Address

BINARY CODED:

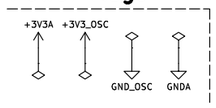
C B A -- VALUE
0 0 0 -- 0
0 0 1 -- 1
0 1 0 -- 2
0 1 1 -- 3
...
1 1 1 -- 7



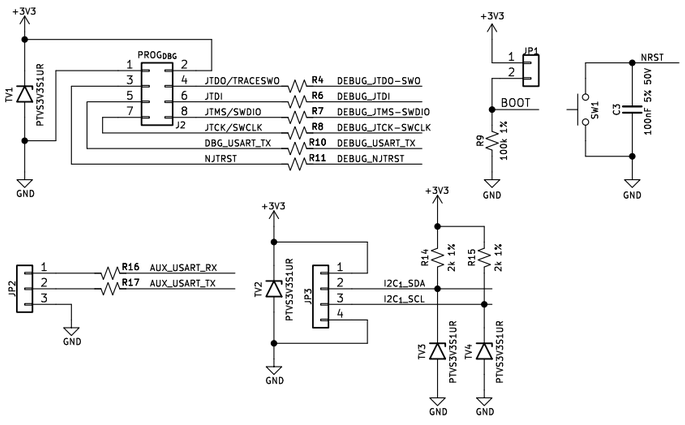
Power supply



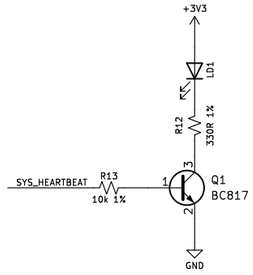
Power flags



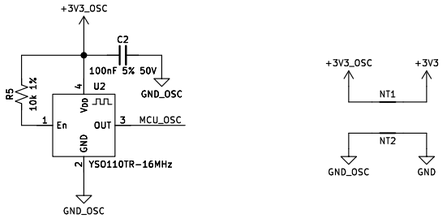
Prog/Debug



Status



Oscillator



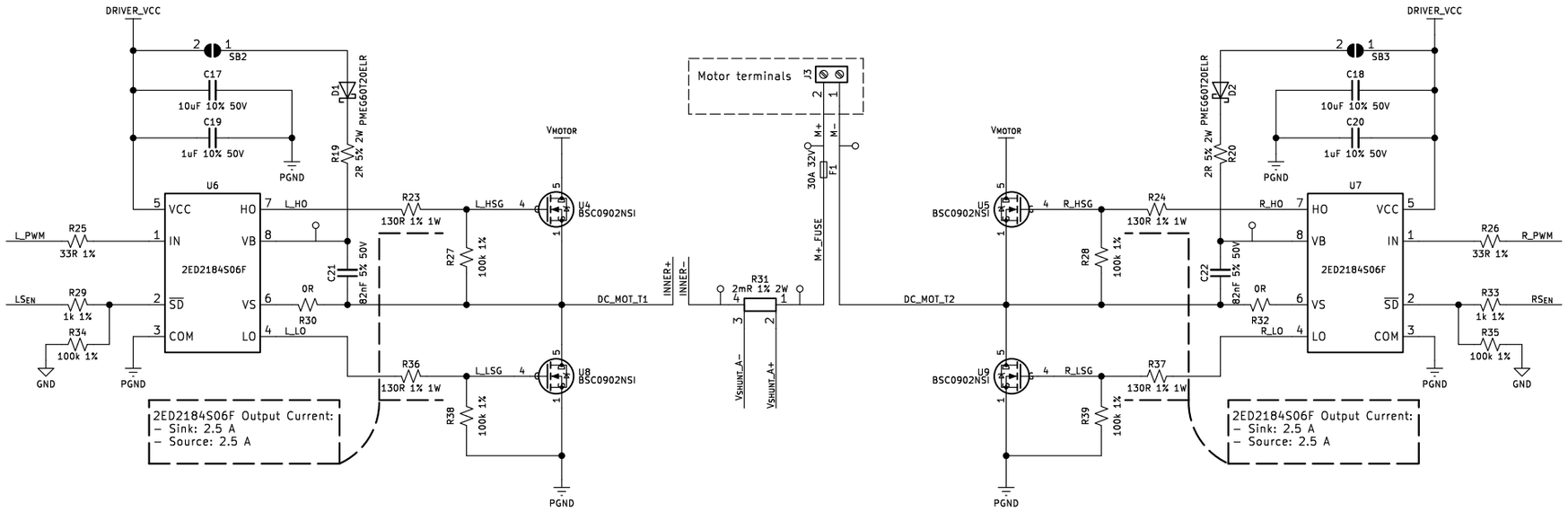
Single Joint Controller Board
Designed by Alves, T. G. (tgiacomellibr)

Sheet: /Control/
File: control.kicad_sch

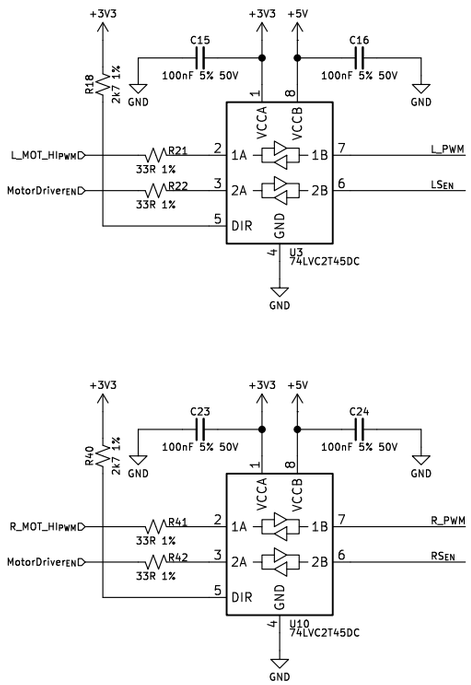
Title: SJC Control

Size: A3	Date: 2024-05-06	**Rev: 1.0**
KiCad E.D.A. 8.0.8		Id: 2/10

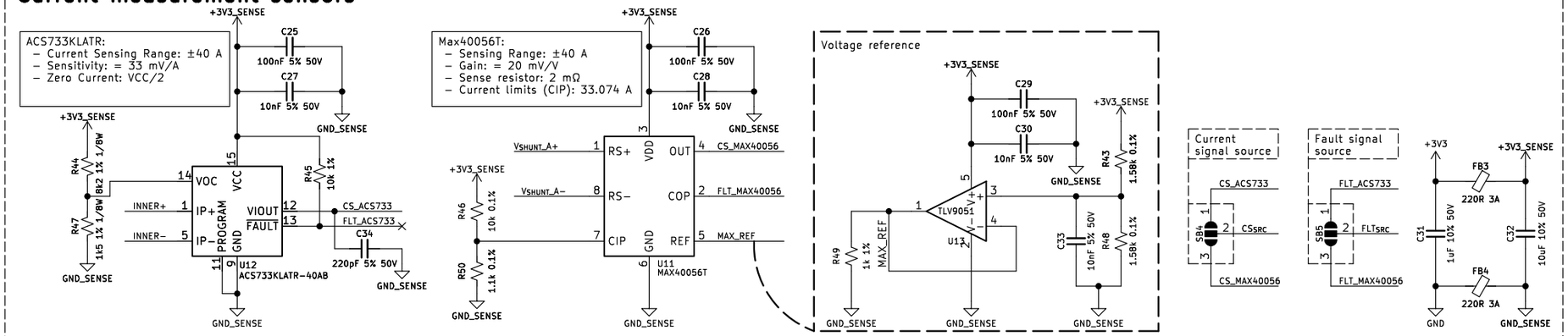
Servomotor interface



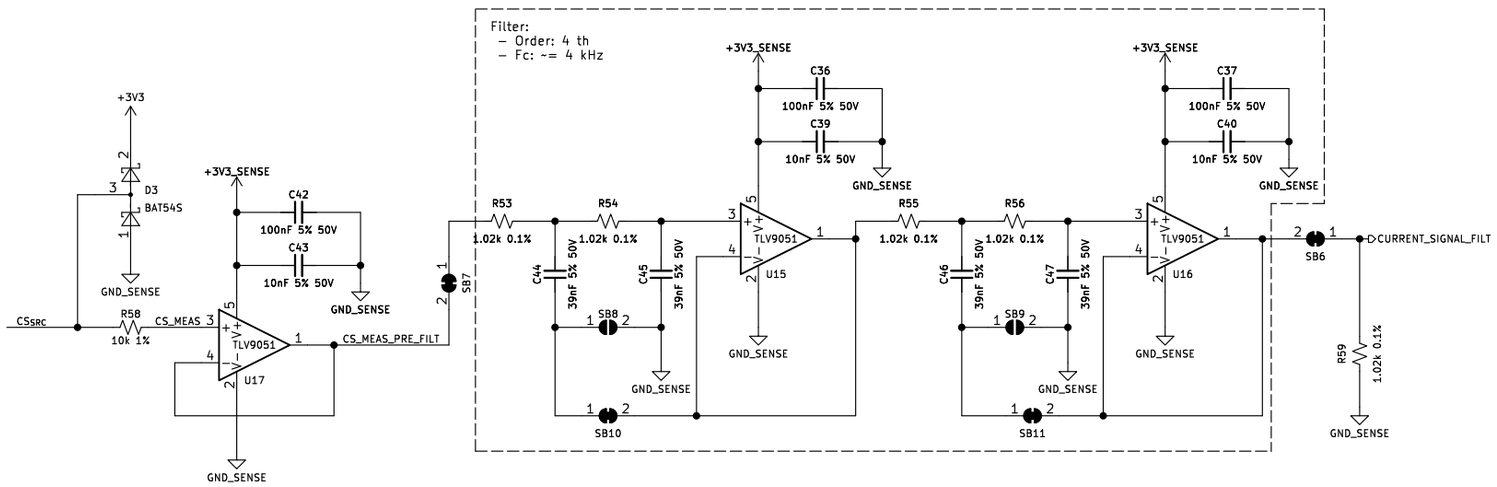
PWM Signals



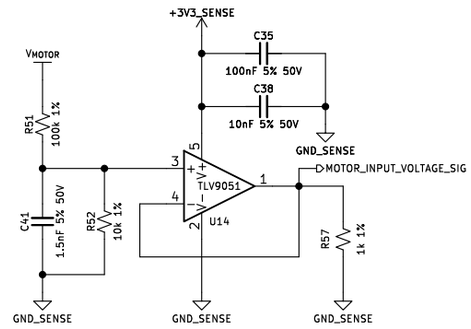
Current measurement sensors



Current signal conditioning interface



H–Bridge input voltage sense



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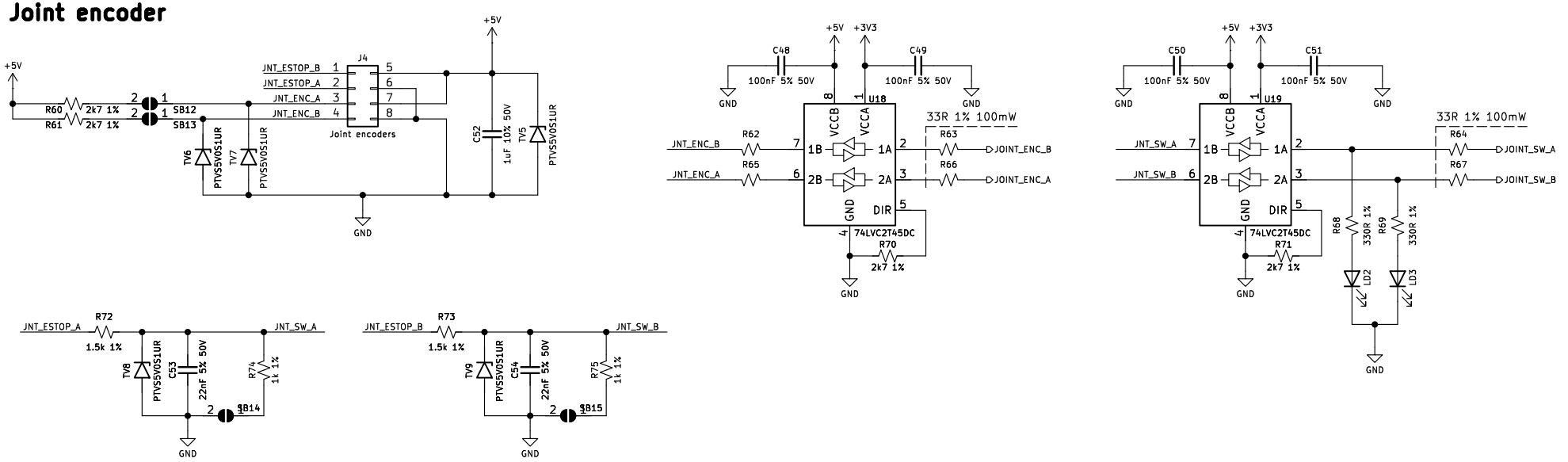
Sheet: /Motor/
File: motor.kicad_sch

Title: SJC Servomotor

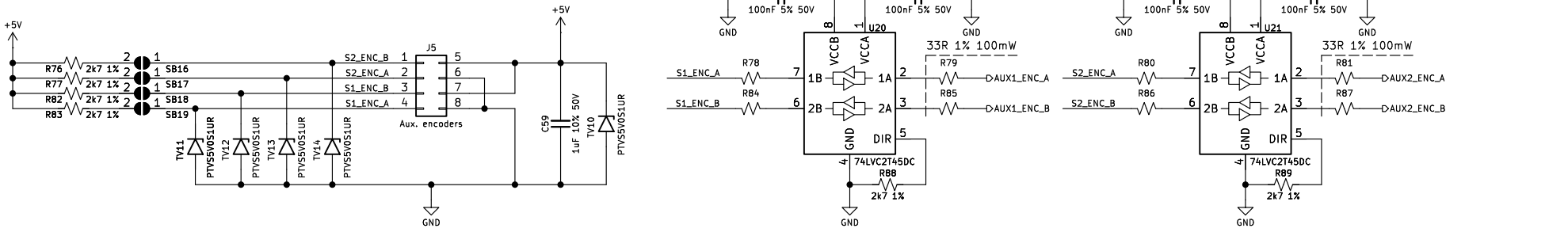
Size: User	Date: 2024-05-06	**Rev: 1.0**
KiCad E.D.A. 8.0.8		Id: 3/10

Encoder interfaces

Joint encoder



Aux encoders



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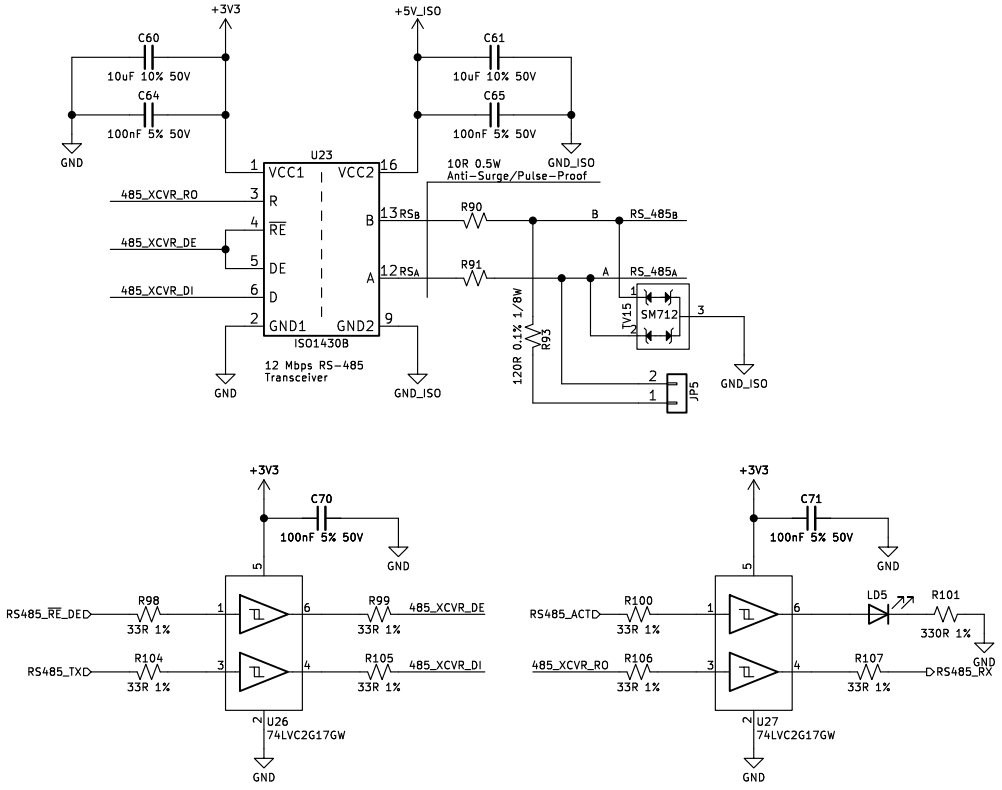
Sheet: /Encoders/
File: encoders.kicad_sch

Title: SJC Encoders

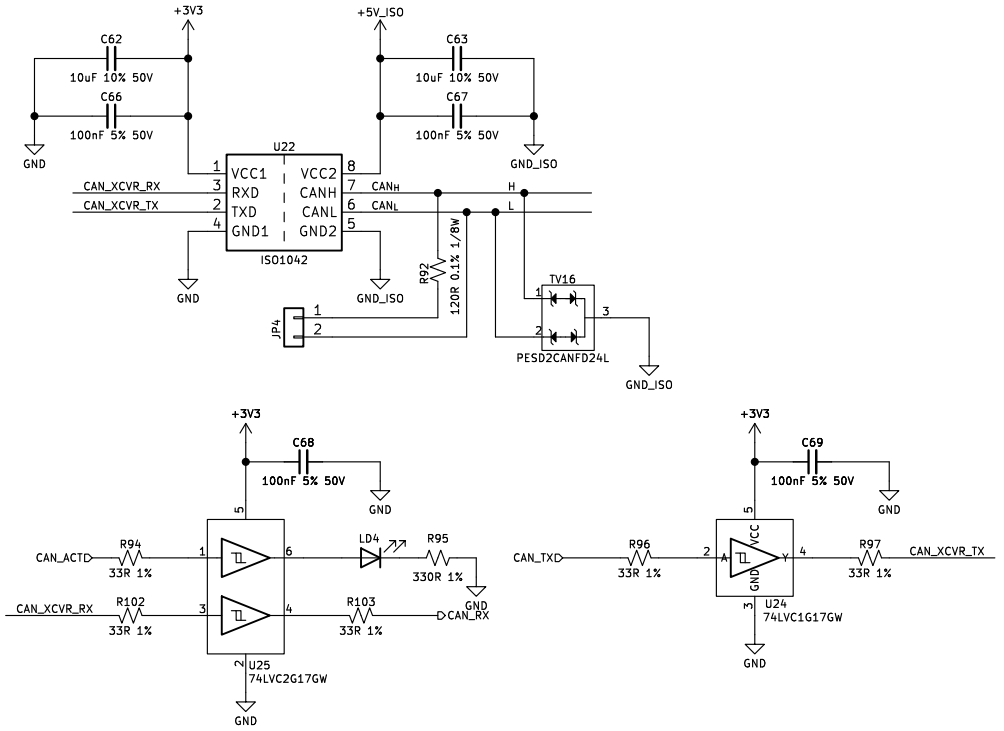
Size: A4	Date: 2024-05-06	**Rev: 1.0**
KiCad E.D.A. 8.0.8		Id: 4/10

Communication

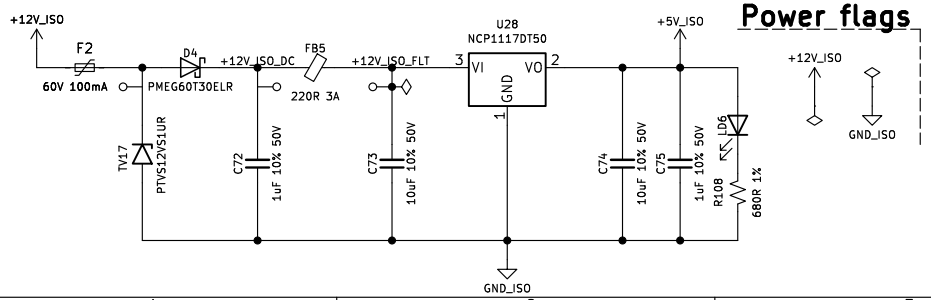
RS-485 (ISO)



CAN (ISO)

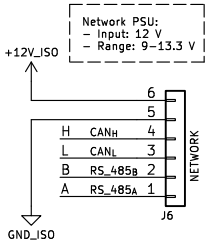


BUS power supply



Power flags

Connector



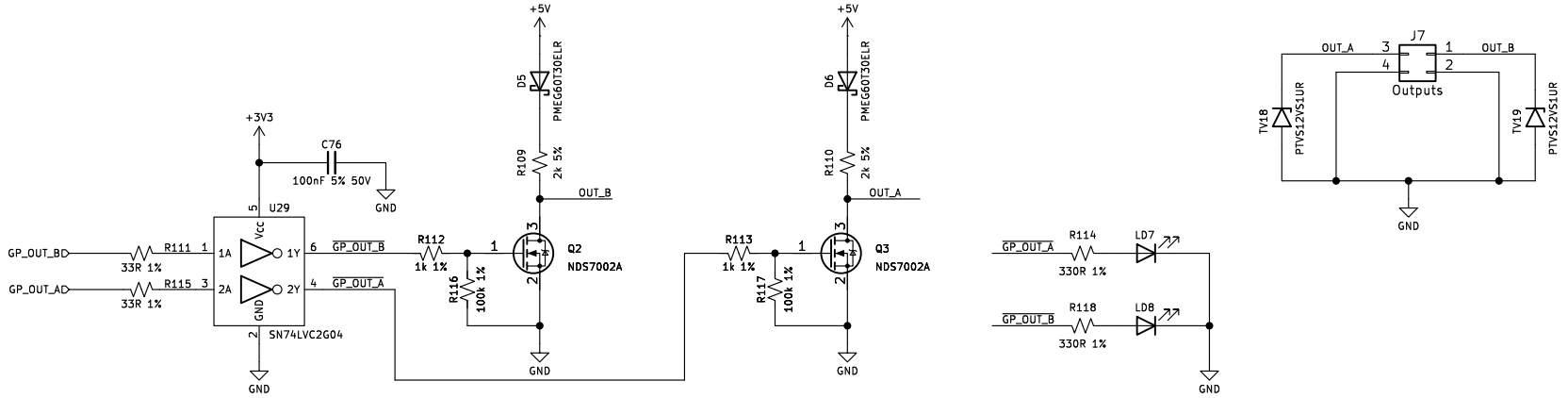
Single Joint Controller Board
Designed by Alves, T. G. (tgiacomellibr)

Sheet: /Network/
File: network.kicad_sch

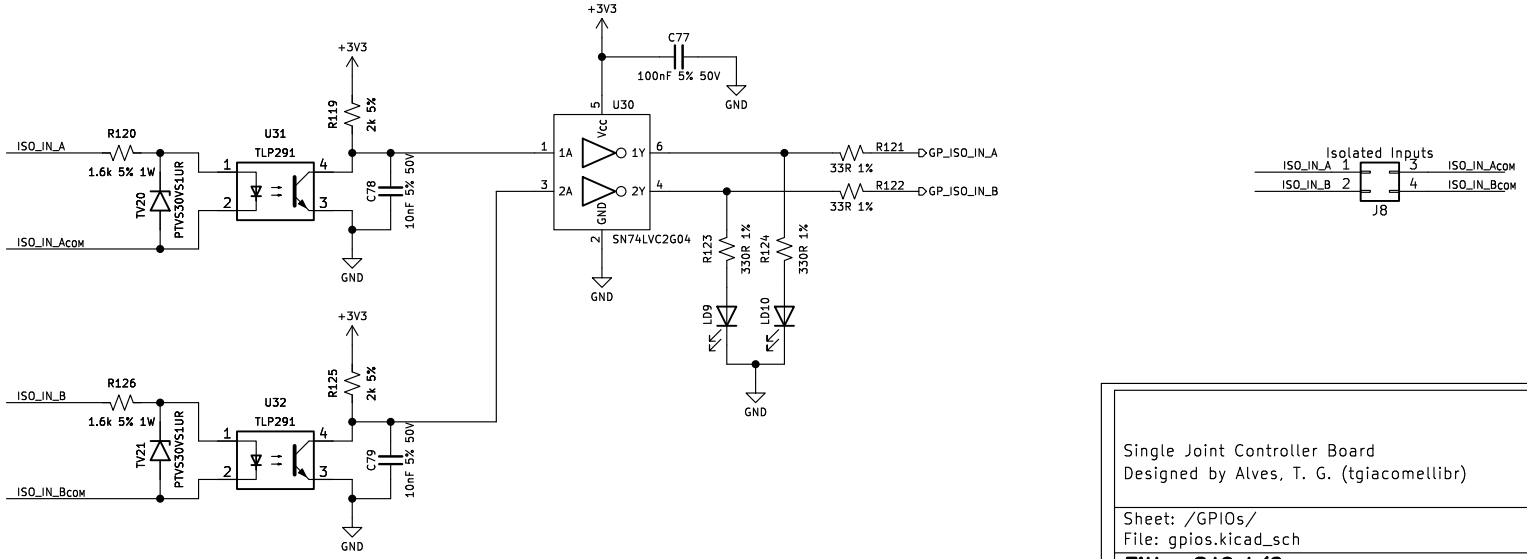
Title: SJC Communication

Size: A4	Date: 2024-05-06	**Rev: 1.0**
KiCad E.D.A. 8.0.8		Id: 5/10

Digital Outputs



Isolated Inputs



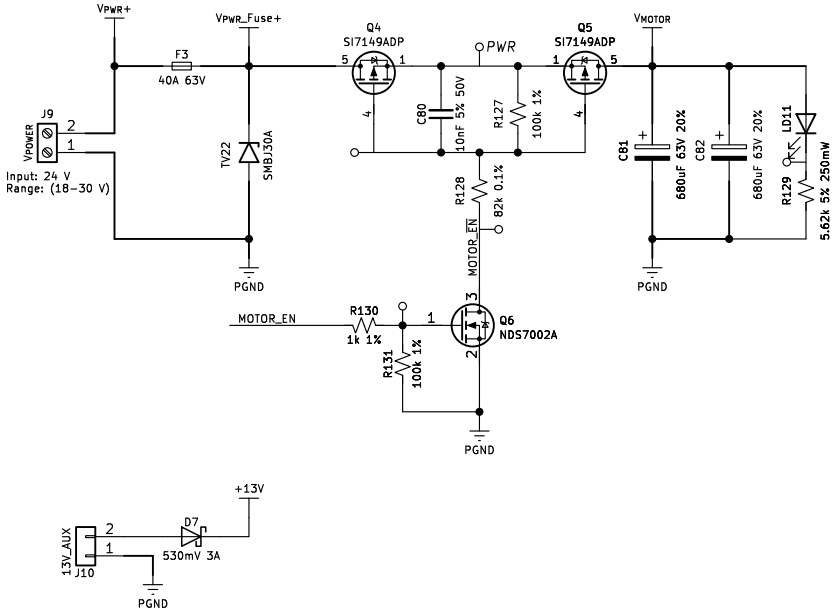
Single Joint Controller Board
Designed by Alves, T. G. (tgiacomellibr)

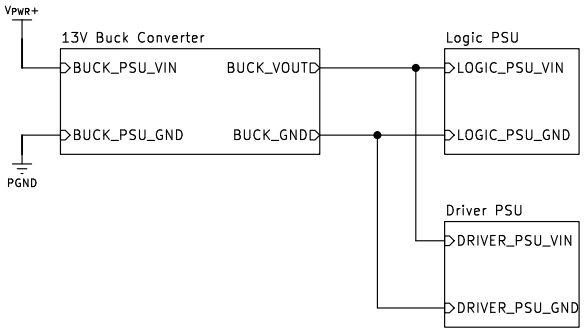
Sheet: /GPIOs/
File: gpios.kicad_sch

Title: SJC I/Os

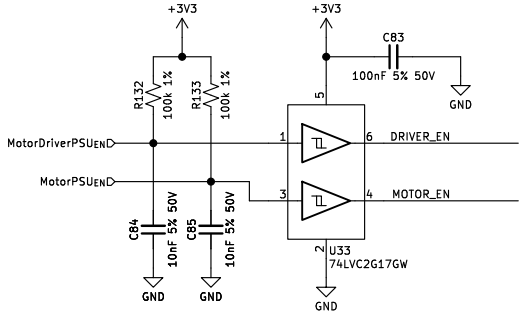
Size: A4	Date: 2024-05-06	**Rev: 1.0**
KiCad E.D.A. 8.0.8		Id: 6/10

Power scheme

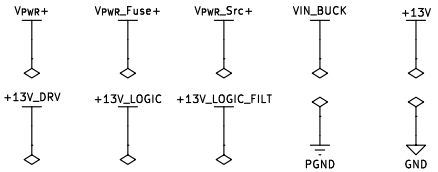




Bypass circuit



Power flags



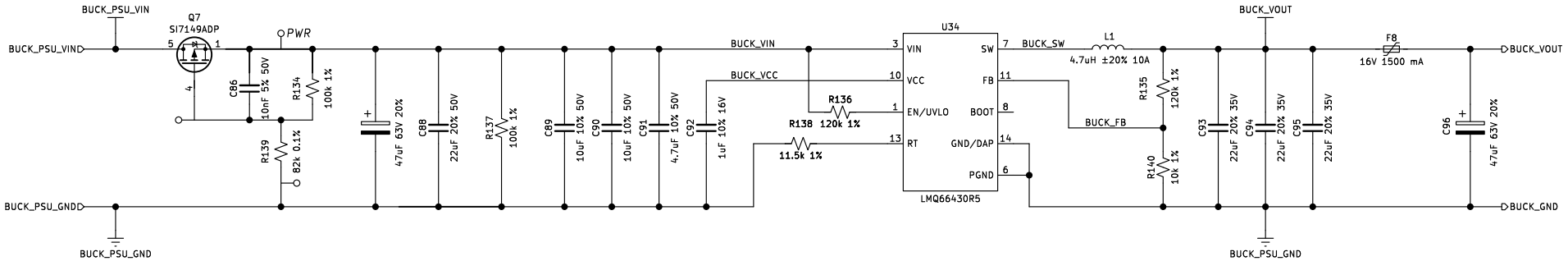
Single Joint Controller Board
Designed by Alves, T. G. (tgiacomellibr)

Sheet: /Power Supply/
File: power_supply.kicad_sch

Title: SJC Power Supply

Size: A4	Date: 2024-05-06	**Rev: 1.0**
KiCad E.D.A. 8.0.8		Id: 7/10

Driver PSU



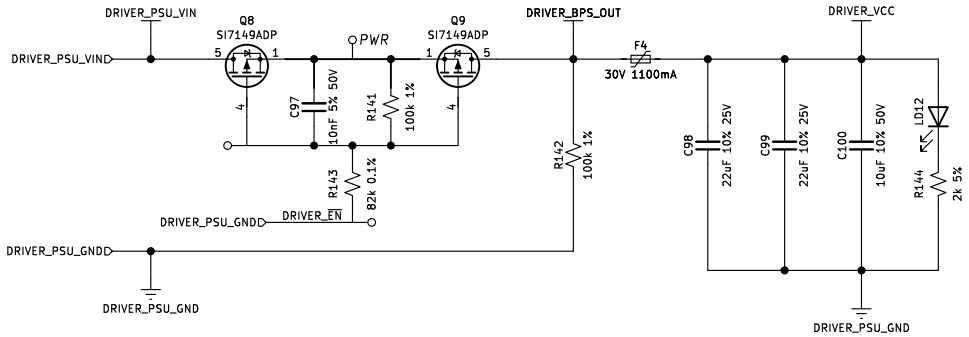
Single Joint Controller Board
Designed by Alves, T. G. (tgiacomellibr)

Sheet: /Power Supply/13V Buck Converter/
File: DC-DC_Converter.kicad_sch

Title:

Size: A4	Date: 2024-05-06	**Rev: 1.0**
KiCad E.D.A. 8.0.8		Id: 8/10

Driver PSU



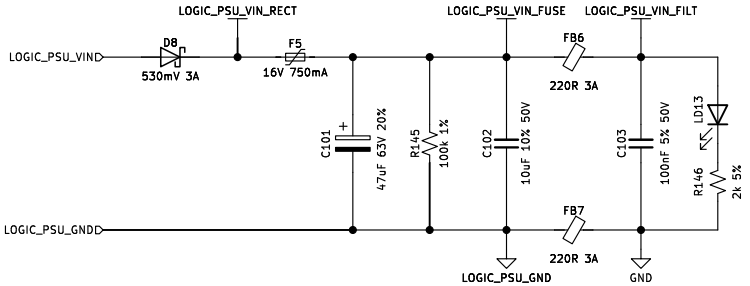
Single Joint Controller Board
Designed by Alves, T. G. (tgiacomellibr)

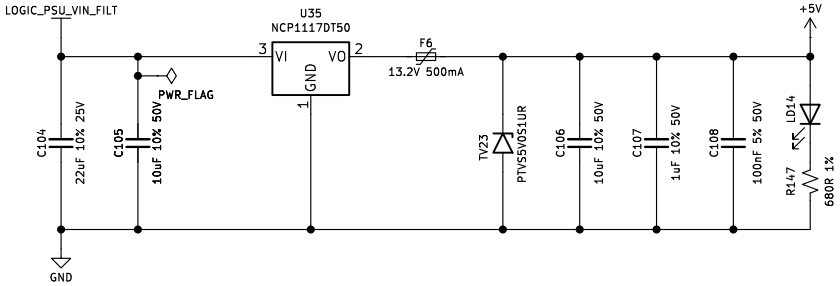
Sheet: /Power Supply/Driver PSU/
File: DRIVER_PSU.kicad_sch

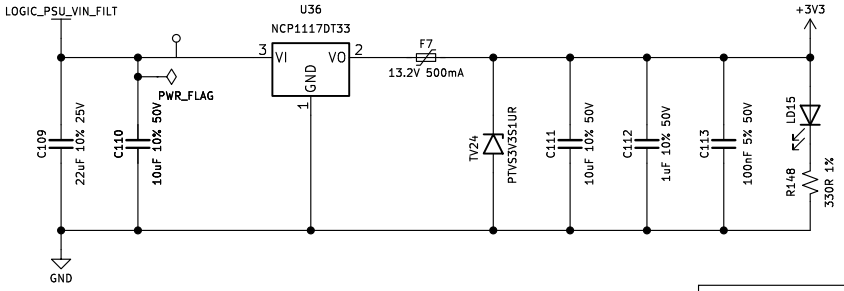
Title:

Size: A4	Date: 2024-05-06	**Rev: 1.0**
KiCad E.D.A. 8.0.8		Id: 10/10

Logic PSU







Single Joint Controller Board
Designed by Alves, T. G. (tgiacomellibr)

Sheet: /Power Supply/Logic PSU/
File: Logic_PSU.kicad_sch

Title:

Size: A4	Date: 2024-05-06	**Rev: 1.0**
KiCad E.D.A. 8.0.8		Id: 11/10